

April Crossword

Kevin Chen, Sohum Uttamchandani

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ONE ACROSS

As spring arrives, a gardener plants tulip bulbs in her flowerbed. She finds that the total number of bulbs N satisfies:

- When arranged in rows of 5, exactly 4 bulbs remain.
- When arranged in rows of 7, exactly 5 bulbs remain.
- When arranged in rows of 9, exactly 8 bulbs remain.

If N is the smallest positive integer satisfying these conditions, find the sum of the digits of N .

ONE DOWN

An artisan is crafting a decorative stained-glass lantern in the shape of a right rectangular prism with a square base. The side length of the square base is 5 inches, and the total surface area of the lantern (including both the top and the bottom faces) is 290 square inches. What is the height of the lantern in inches?

TWO DOWN

A starship navigator is plotting a hyperspace route on a 2D galactic coordinate grid. The ship starts at the origin $(0, 0)$ and must reach a space station located at coordinates $(11, 2)$. To navigate safely through an asteroid field, the ship's autopilot restricts all movements to jumps of exactly one unit East or one unit North. How many unique valid routes can the navigator plot to reach the station?

THREE ACROSS

April has 30 days. Let $S \subseteq \{1, 2, \dots, 30\}$ be the set of rainy dates in April, with $|S| = n$. Suppose:

- $\sum_{k \in S} (k - 8)^2 = 122n$.
- $\sum_{k \in S} (k + 6)^2 = 486n$.

Find twice the mean of S .